

AAKASH KHARB

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EDUCATION

Maharshi Dayanand University, Rohtak

Bachelor of Technology (B.Tech) in Computer Science Engineering (AI & ML) | Oct 2022 – Jun 2026

CGPA: 3.78 / 4.00 (WES evaluated)

Capstone Project: ChickpeaOmicsExplorer : ML-assisted transcriptomics platform

RESEARCH & PROFESSIONAL EXPERIENCE

Center for Biotechnology, Maharshi Dayanand University

Research Engineer – AI/ML (Bioinformatics) | Oct 2024 – Present

- Designed and deployed ChickpeaOmicsExplorer, an ML-assisted transcriptomics and multi-omics exploration platform for Cicer arietinum.
- Built machine learning pipelines using SVM, feature engineering, and hyperparameter tuning, achieving robust accuracy on gene expression prediction tasks (housekeeping vs non-housekeeping genes).
- Developed interactive Django dashboards and deployed scalable, cloud-native infrastructure on Google Cloud Platform (GCP).
- Integrated heterogeneous biological datasets, automated ETL pipelines, and optimized backend databases for real-time genomic analysis.

TGID (Overseas Education Consulting Firm), New Delhi

Software Engineer Associate | Nov 2024 – Mar 2026

- Developed and maintained internal platforms and automation tools supporting counseling, application tracking, and CRM workflows.
- Built scripts, APIs, and database workflows to automate student onboarding, university matching, and documentation pipelines.

TEACHING & ACADEMIC SERVICE

StudentTechHub, Maharshi Dayanand University

AI Mentor | Jan 2023 – Jun 2026

- Conduct weekly hands-on sessions on AI, Machine Learning, Cloud Computing, and Data Science, combining theory with live coding and mini-projects.
- Mentored students on Scikit-learn, TensorFlow, Streamlit, project development, code reviews, and research-oriented career pathways.

Department of Computer Science & Engineering, MDU

Teaching Assistant / Laboratory Assistant | Feb 2024 – Nov 2025

- Assisted laboratory courses in Machine Learning, Neural Networks, Deep Learning, and Big Data Analytics using Python, MATLAB, and Hadoop.
- Supported students in regression, classification, CNNs, RNNs, data visualization, debugging, and assignment evaluation.

PUBLISHED RESEARCH PAPERS

Kharb, A. *Explainable AI in Transplant Medicine: Transforming Kidney Donor Matching and Graft Outcome Prediction.*

International Journal of Research in Medical Science & Technology (IJRMST), Vol. 16, Jul–Dec 2023, DOI: 10.37648/ijrmst.v16i01.018.

Kharb, A. *Benchmarking CNN, LSTM, GRU and Transformer Models for Twitter Sentiment Analysis.*

International Journal of Universal Science & Engineering (IJUSE), Vol. 11, Issue 1, Jan–Dec 2025; e-ISSN: 2454-759X | p-ISSN: 2454-7581

Kharb, A. *A Survey of Big Data Approaches in Natural Language Formation and Computing Science.*

International Journal of Research in Science and Technology (IJRST), Vol. 15, Issue 1, 2025, DOI: 10.37648/ijrst.v15i01.007.

Kharb, A. *Generative AI for Risk Assessments: A Taxonomy Across Models, Lifecycles and Domains.*

International Journal of Technology, Science and Engineering (IJTSE), Vol. 7, Issue IV, Oct–Dec 2024; ISSN: 2457-1016

Kharb, A. *Developing a Smart, Integrated Irrigation System Using ML and IoT with KNN.*

International Journal of Technology, Science and Engineering (IJTSE), Vol. 7, Issue I, Jan–Mar 2024; ISSN: 2457-1016.

Manuscript Under Review:

ChickpeaOmicsExplorer (COE): An ML-assisted Multi-Omics Platform for Functional Genomics. Journal for Advanced research

Deciphering the synergistic role of Serendipita indica and salicylic acid in arsenic stress tolerance of Ocimum sanctum using physio-biochemical analyses and machine learning

SELECTED ACADEMIC PROJECTS

KIDS-NIDS: AI-Powered Neonatal Care Assistant

- Designed a dual-tier system comprising a conversational AI for parents and a clinical decision-support interface for healthcare professionals, integrating document intelligence and real-time decision workflows.

TechWill × Olympics: Paris 2024 Data & Insight Platform

- Built a cloud-integrated analytics and visualization platform with an AI chatbot (Gemini API) for real-time Olympic data exploration.

Game Recommendation System (Content-Based)

- Implemented KNN and cosine similarity models on Steam data; developed an interactive Streamlit application with cloud-hosted models.

TechWill × Docx: Intelligent PDF Summarization Platform

- Developed an AI-driven system for real-time PDF summarization, conversational Q&A, and visual analytics using NLP and cloud storage.

TECHNICAL SKILLS

- **Programming:** Python, R, MATLAB, SQL, C/C++, Java, JavaScript
- **Machine Learning & AI:** Supervised/Unsupervised Learning, Deep Learning, NLP, Computer Vision, Recommender Systems, Explainable AI
- **Frameworks & Libraries:** Scikit-learn, TensorFlow, PyTorch, NumPy, Pandas, SciPy, Matplotlib
- **Cloud & MLOps:** Google Cloud Platform, Microsoft Azure, Vertex AI, Azure AI Studio; model deployment, CI/CD (basic), Docker (basic)
- **Data & Systems:** MySQL, PostgreSQL, MongoDB; ETL pipelines, cloud-native Streamlit applications
- **Tools:** Git, GitHub, VS Code, Jupyter Notebook, Google Colab
- **Operating Systems:** Linux, macOS, Windows

AWARDS & HONOURS

- **Research Gem Award** — Maharshi Dayanand University (2024)
- **Chancellor's Recognition Award** — Top 10% of department (2024)
- **Top 1% Batch Brilliance Award** — Academic excellence across four consecutive semesters
- **Melton Foundation Fellowship** — Selected among five fellows from India (International)
- **International Award for Young People (IAYP)** — Gold Award Recipient

LEADERSHIP & COMMUNITY ENGAGEMENT

- **IEEE MDU Student Branch** — Active Member & Volunteer
- **Training & Placement Cell, MDU** — Placement Coordinator
- **Youth Red Cross MDU** — Volunteer & Event Coordinator
- **Nehru Yuva Kendra (Govt. of India)** — National Service Volunteer
- **NITI Aayog & Piramal Foundation** — Flood-relief and senior-citizen welfare volunteer